STUDY SHEET · FINANCIAL ANALYSIS

Consolidated financial statements

Groups, full consolidation, goodwill and NCI, intragroup transactions and the equity method, with every formula and a worked number for each.

— Tipoki

CONTENTS

01 Groups and types of investment

02 Goodwill and NCI at acquisition

03 Consolidating after the acquisition

04 Internal transactions

05 The equity method

06 Worked cases to know

Slides parts 1 to 4 · Assignments 1 to 5 · Cases 1 and 2 (IFRS)

> *Consolidated statements show a group **as if it were one single entity**. The whole topic comes down to one question, asked three times: **how much of this entity do we control, and what must be cancelled so that nothing is counted twice?***

01 Groups and types of investment

The percentage of voting rights decides the accounting method.

Under IFRS a **group** is a parent and all its subsidiaries. As soon as a parent owns one subsidiary, it must prepare **consolidated financial statements (CFS)**. Two exemptions: the parent is itself a wholly-owned subsidiary, or its debt and equity are not traded on a public market.

Stake	Relationship	Method
Above 50% of votes (or control by agreement, statutes, board power)	**Subsidiary** (control)	Full consolidation
20% to 50%, significant influence	**Associate**	Equity method
Joint control, unanimous consent on key decisions	**Joint venture**	Equity method
Below 20%, no significant influence	Simple investment (IFRS 9)	No consolidation

- **Control** can exist at or below 50%: a casting vote on the board, or an agreement adding voting rights (45% + 8% = 53%).
- Owning more than 50% is not enough if control cannot actually be exercised: that is then only significant influence.
- The strongest evidence of significant influence is **representation on the board**.

02 Goodwill and NCI at acquisition

Eliminate the investment against the subsidiary's equity, then see what is left over.

Goodwill	Price paid minus the parent's share of the **fair-valued** equity of the subsidiary at the acquisition date. Not amortised: it is tested for impairment.
Bargain purchase (badwill)	Negative goodwill: recognised as a **gain** in the consolidated income statement.
NCI (non-controlling interests)	The share of the subsidiary's equity not owned by the parent. Shown in equity, on its own line.

AT THE ACQUISITION DATE

$$Goodwill = Price - \%\ owned \times Equity_S$$

$$NCI = (100\% - \%\ owned) \times Equity_S$$

$Equity_S$ = share capital + all reserves, after fair value adjustments

WORKED EXAMPLE · GOODWILL AND NCI

Statement. A buys **80%** of B for **7,200**. B's capital is 2,000, legal reserves 2,500 and retained earnings 1,400.

Equity of B = 2,000 + 2,500 + 1,400 = 5,900.

Group share = 5,900 × 0.8 = 4,720, so **goodwill** = 7,200 − 4,720 = **2,480**.

NCI = 5,900 × 0.2 = **1,180**.

Variant. If a non-current asset of B is undervalued by 150, equity at fair value is 6,050: goodwill = 7,200 − 4,840 = **2,360** and NCI = 1,210.

Bargain purchase. P buys 75% of S for 2,450 when S's equity is 2,000 + 1,900 = 3,900: group share 2,925, so 2,450 − 2,925 = **−475**, a gain.

KEY TAKEAWAYS

- 100% of the subsidiary's equity is eliminated, whatever the percentage owned.
- If the price is below the group share, the difference is a gain, not goodwill.
- Adjustments are made on the consolidation worksheet only: the individual companies' books do not change.

03 Consolidating after the acquisition

Goodwill stays put; reserves and NCI move with the subsidiary's equity.

AFTER ACQUISITION

$$Group\ reserves = \%\ owned \times (Equity_S\ now - at\ acquisition)$$

$$NCI = (100\% - \%\ owned) \times Equity_S\ now$$

Goodwill does not change with later changes in the subsidiary's equity

WORKED EXAMPLE · AFTER THE ACQUISITION

Statement. P bought **75%** of S for **2,450** when S's equity was capital 2,000 + retained earnings 1,200 = 3,200. S's retained earnings are now 1,900.

Goodwill (fixed at acquisition) = 2,450 − 3,200 × 0.75 = **50**.

Group reserves = (1,900 − 1,200) × 0.75 = **525**.

NCI = (2,000 + 1,900) × 0.25 = **975**.

Falling equity (Planet and Sun). Planet buys 80% of Sun for 160 when Sun's equity is 180 (100 + 10 + 20 + 50): goodwill = 160 − 144 = **16**. Sun's equity is now 160, so group reserves = (160 − 180) × 0.8 = **−16** and NCI = 160 × 0.2 = **32**.

A full consolidation **adds 100%** of the subsidiary's assets, liabilities, revenues and expenses line by line, even the part financed by NCI. A subsidiary acquired during the year only contributes its **post-acquisition** income.

04 Internal transactions

What the group does with itself must disappear from the group's statements.

Three families must be eliminated: **intragroup sales and purchases**, **intragroup balances** (receivables and payables, loans and interest), and **unrealised profit** on goods or assets still held inside the group.

Situation	Adjustment
Sale of goods inside the group	Cancel sales against purchases (COGS). Remove the unrealised profit from inventories and from the seller's income, in proportion to the goods still held.
Receivable / payable, loan, interest	Cancel both sides; no profit effect. Interest revenue and expense cancel each other (350,000 at 5% for 10 months = 14,583).
Sale of a depreciable asset	Remove the unrealised profit from the asset, then **reverse the excess depreciation** it generated.

UNREALISED PROFIT ON GOODS

$$\textit{Unrealised profit} = \textit{profit} \times \textit{share of goods still in the group}$$

profit 150, 40% still in stock: 150 × 40% = 60

WORKED EXAMPLE · INTRAGROUP TRANSACTIONS

Goods. A sells inventory to B for 500 with a profit of 150; 60% has been sold to third parties. Unrealised profit = 150 × 40% = **60**.

Machinery. On 1 Jan X1 B sells a machine to A for 120,000, making a profit of 60,000. Remaining life: 5 years.

Excess depreciation = 60,000 / 5 = **12,000** a year.

Net adjustment at 31 Dec X1 = 60,000 − 12,000 = **48,000** (machinery −60,000, accumulated depreciation +12,000).

Loan. A lends 350,000 to B on 1 March at 5%. For the 10 months to 31 December, interest revenue and expense of 350,000 × 5% × 10 / 12 = **14,583.33** are eliminated; the interest accrued at year-end (1 month) is 1,458.33.

KEY TAKEAWAYS

- If the **subsidiary** is the seller, the unrealised profit also changes NCI and group reserves.
- Goods already sold to third parties have realised their profit: nothing to remove.
- Balances can only be eliminated if both companies agree on them.

05 The equity method

Associates and joint ventures: the investment is updated, never consolidated line by line.

With the equity method, **no assets or liabilities of the associate are added**. The investment is carried forward in one line, "Equity investment in A", and "tipped up" each year by the investor's share of the associate's post-acquisition earnings (reduced by any impairment).

EQUITY INVESTMENT

$$Investment = cost + \%\ owned \times (Equity_A\ now - at\ acquisition)$$

Goodwill is computed as for a subsidiary and stays included in the carrying value

WORKED EXAMPLE · EQUITY METHOD

Statement. P buys **40%** of A for **340,000**. At acquisition A has capital 300,000 and retained earnings 500,000, so equity is 800,000. Retained earnings are later 800,000.

Goodwill = 340,000 − 0.4 × 800,000 = **20,000**.

Post-acquisition profit = (800,000 − 500,000) × 0.4 = **120,000**, credited to group reserves.

Investment on the balance sheet = 340,000 + 120,000 = **460,000** (goodwill included).

KEY TAKEAWAYS

- Subsidiary: the investment is **replaced** by net assets. Associate: the investment is **carried forward** and updated.
- A group needs at least one subsidiary to exist, and therefore to consolidate.

06 Worked cases to know

The two cases from the course, with the givens needed to redo them.

CASE 1 · A / B / C

Statement. A buys **80%** of B for **450,000** when B's equity is 420,000. B's equity at the consolidation date is 394,000. A also owns 25% of C, whose equity rose from 50,000 to 190,000.

Goodwill = 450,000 − 0.8 × 420,000 = **114,000**.

Group reserves from B = (394,000 − 420,000) × 0.8 = **−20,800**.

C (associate): share of the post-acquisition change = (190,000 − 50,000) × 25% = **35,000**, added to the investment.

CASE 2 · PUSH / SHOVE

Statement. Push buys **75%** (150,000 of 200,000 shares) of Shove for **295** (thousand €) on 1 Sep X1. At that date Shove has capital 200, retained earnings 60 and PPE undervalued by 100. At 31 Aug X3 Shove has capital 200, retained earnings 140 and FV adjustment 100.

Goodwill = 295 − 0.75 × (200 + 60 + 100) = 295 − 270 = **25**.

NCI = (200 + 140 + 100) × 25% = **110**.

Intragroup items, all sold by Push: goods bought at 100 and sold for 125, 4/5 still in stock, so unrealised profit = 25 × 4/5 = **20**; machine sold for 198 with a profit of 18 and 10 years left, so excess depreciation = **1.8** and net reduction = 18 − 1.8 = **16.2**; loan of 130 eliminated (50 remains external).

Consolidated retained earnings = 405 − 20 − 16.2 − 140 = **228.8**.

Pescanova / Pescamar: Pescanova owns 60% of Pescamar but political problems prevent it from exercising control while it keeps some influence. Majority ownership without the ability to control gives only **significant influence**: Pescamar is an associate (equity method).

EXAM ROUTINE

- 1. Classify each stake: subsidiary, associate, joint venture or simple investment.
- 2. Compute goodwill and NCI at the acquisition date.
- 3. Update reserves and NCI for the post-acquisition change in equity.
- 4. Eliminate sales, balances and unrealised profit (and excess depreciation).
- 5. Add the associate's updated investment, then check that the balance sheet still balances.
